

DEVELOPMENT OF AN R PACKAGE (aLBI) FOR ESTIMATING LIFE HISTORY PARAMETERS AND STOCK STATUS FROM LENGTH FREQUENCY DATA

Ataher Ali^{1*} (ID: 18207076), Dr. Mohammed Shahidul Alam¹

¹Department of Fisheries, University of Chittagong, Chattogram-4331, Bangladesh



MS THESIS DEFENCE - 2022
DEPARTMENT OF FISHERIES
UNIVERSITY OF CHITTAGONG

ABSTRACT

Stock assessment is a difficult task for sustainable fisheries management especially when population and fisheries data are limited. This study aims to develop an accessible and tested R package (aLBI) that applies length frequency and provides functions for estimating length-based indicators (LBIs) to measure stock health. The primary objective is to streamline the calculation and visualization of the estimated metrics for a clear understanding. The package includes three main functions, i.e., FishPar (for calculating key LBIs such as maximum length (L_{max}), asymptotic length (L_{∞}), length at sexual maturity (L_{mat}), optimum length (L_{opt}), $L_{opt} \pm 10\%$ (L_{opt_p10} , L_{opt_m10}), and percentages of fish at maturity (P_{mat}), optimum length (P_{opt}), and mega spawners (P_{mega}) with confidence intervals), FishSS (for evaluating stock biomass status), and LWR (for visualizing length-weight relationship). This study also evaluated the performance of the package by applying length frequency data of *Sperata aor* collected from Kaptai Lake. The findings with a calculated optimum bin size of three (cm) demonstrate its utility in producing acceptable stock assessments, with mean estimates

for L_{max} , L_{∞} , L_{mat} , and L_{opt} at 78.01 cm, 82.12 cm, 43.73 cm, and 47.03 cm, respectively, as well as respective confidence intervals. Besides, percentages of sexually mature fish, optimum length, and mega spawners are 2.04%, 2.38%, and 0.35%, against targets of 100%, 100%, and 20% respectively. By integrating advanced statistical methods and user-friendly interfaces, aLBI enables effective monitoring of fish populations and informed decision-making. Tested on various bin size (1–5 cm) with the collected length frequency data, the package performed consistently. Future recommendations include expanding functionality to cover more species and integrating additional biological parameters for enhanced stock assessments. Continued collaboration with the fisheries science community will ensure aLBI remains a cutting-edge tool for sustainable fisheries practices.

Key words: R package development, stock assessment, aLBI, length-based indicators (LBIs), target biomass, limit biomass, Kaptai Lake.

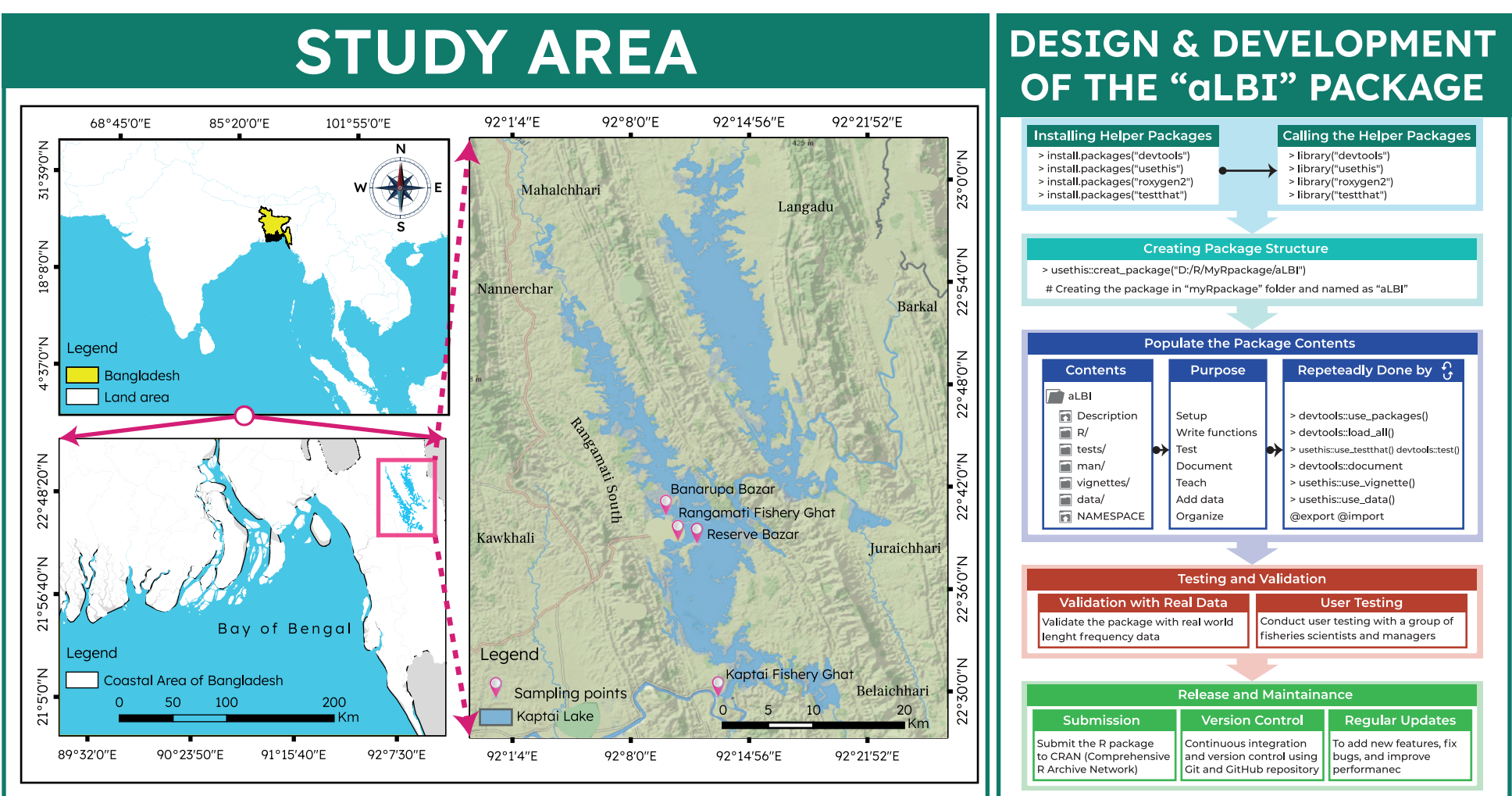
INTRODUCTION

Effective fisheries management is crucial to address the global issue of over-exploited fish stocks. Traditional stock assessments, which require extensive, high-quality data, are often difficult to apply in data-limited regions. To overcome these challenges, scientists have developed alternative methods that rely on more readily available length-frequency data. One such approach, based on Froese's length-based indicators (LBIs), provides a simple yet effective way to evaluate a stock's health. This method helps determine if a stock is experiencing growth or recruitment overfishing and whether its spawning biomass is at a sustainable level. To make this methodology more accessible, a new R package, "aLBI," is being developed to automate the estimation of these indicators, providing a valuable and user-friendly tool for fisheries researchers and managers. This package also incorporates a decision-tree to assess stock status against biological reference points, making it a comprehensive tool for management.

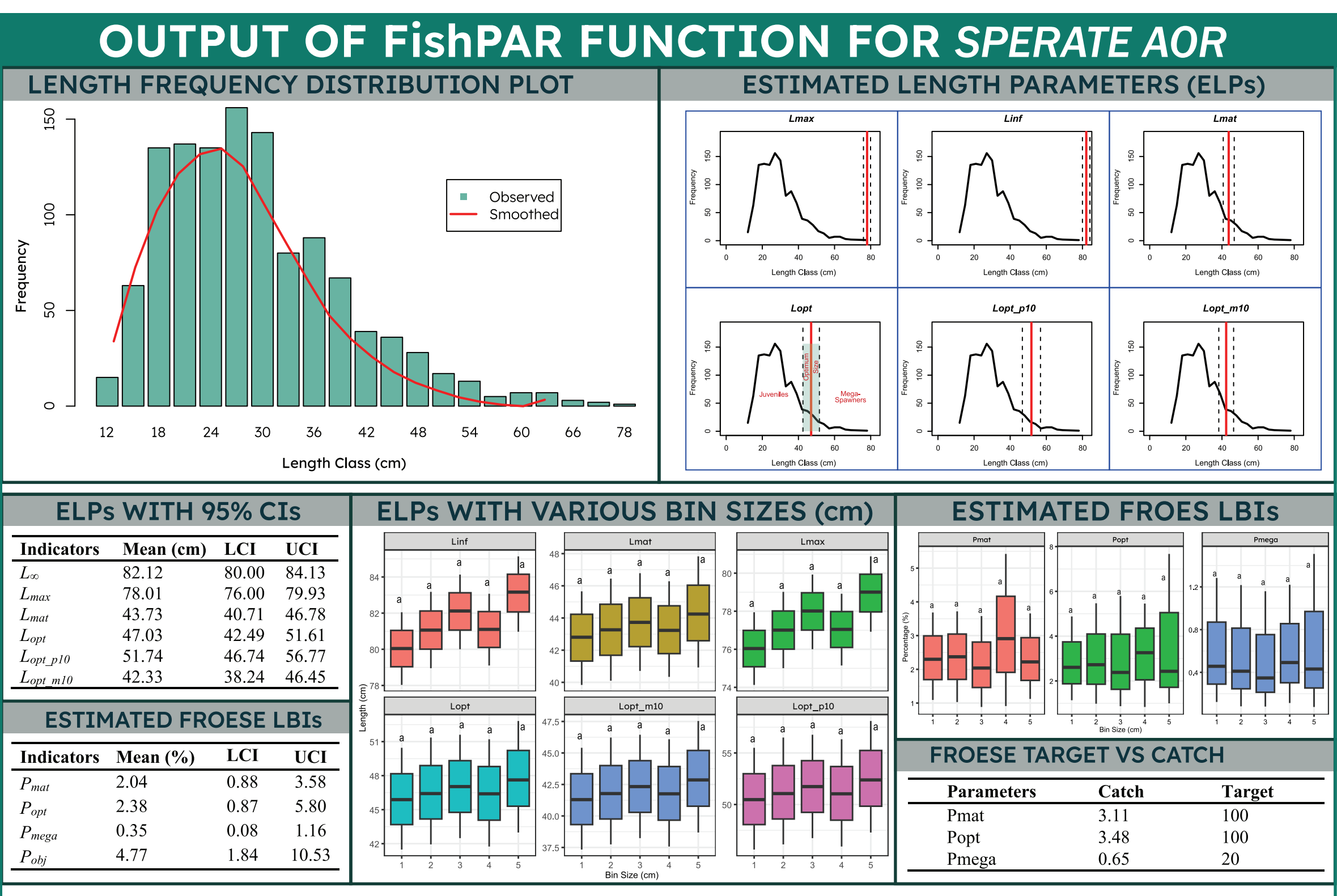
OBJECTIVES

- To develop a flexible R package with core functions that estimate life history parameters Froese's length-based indicators (LBIs) from length-frequency data.
- To validate the functionality and accuracy of the "aLBI" package by applying it to a real-world dataset of *S. aor* from Kaptai Lake.

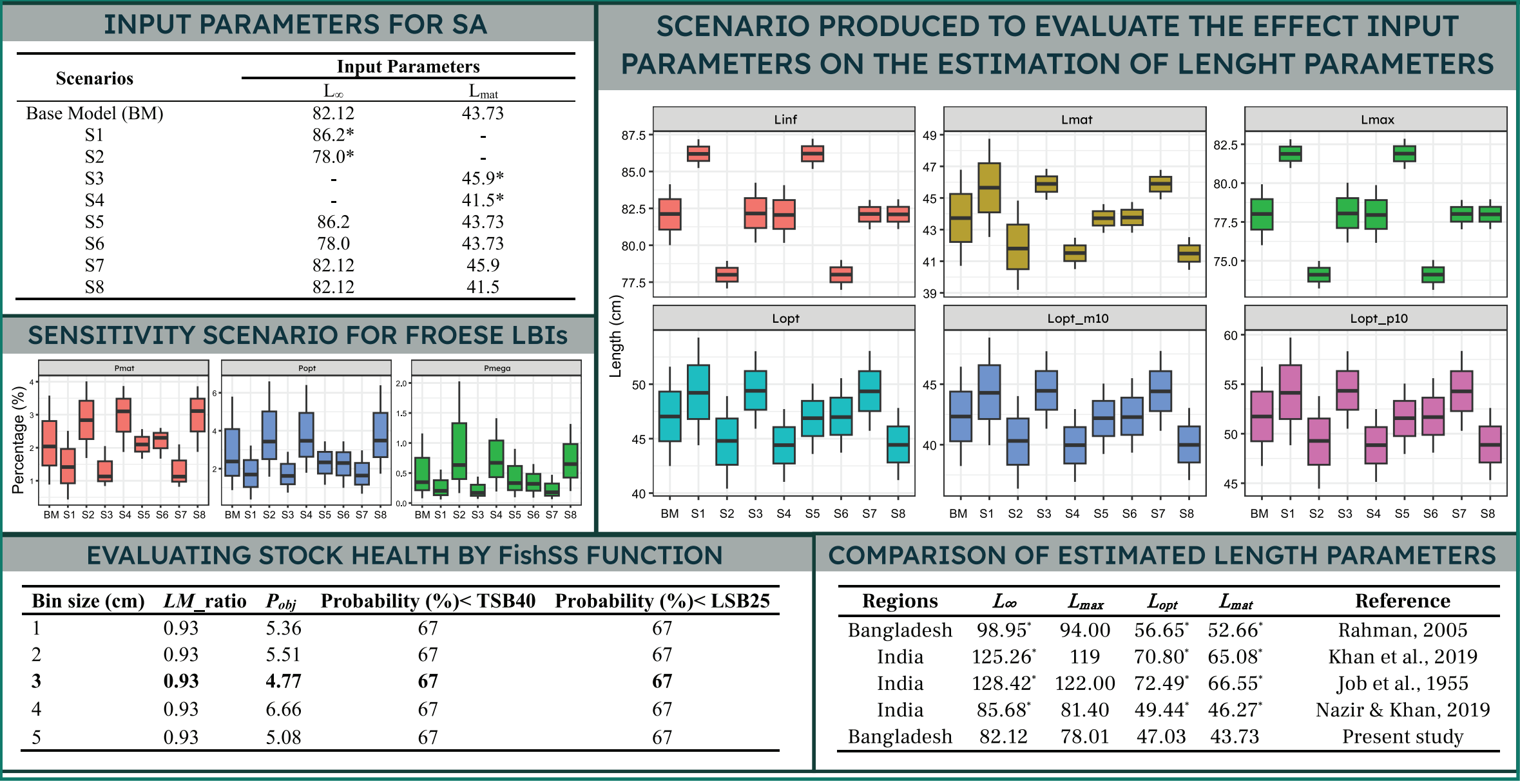
METHODOLOGY



RESULTS



SENSITIVITY ANALYSIS (SA) FOR ESTIMATED PARAMETERS



CONCLUSION

This study developed and utilized a new R package named 'aLBI' to provide more reliable and robust estimates of length-based indicators (LBIs) for assessing fisheries sustainability. The package was specifically applied to evaluate the stock status of *S. aor* from Kaptai Lake in Chittagong, Bangladesh. The results, which were found to be consistent regardless of the bin sizes applied, revealed that this fishery is suffering from both growth and recruitment overfishing. Furthermore, the outputs from the 'aLBI' package indicated that the current spawning biomass of the stock is below the established reference points. To protect juvenile fish and ensure the long-term sustainability of the fishery, the study recommends a lower length limit of 47 cm, which was identified as the optimum length for this specific species.

SCAN HERE TO FIND ALL THE REFERENCES

